Paper 75: z Probed at Deep P; Susceptibility γ Null;

FSS Blocking and Hyperscaling Violation in the Zone-Mean VCML Phase

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Abstract

We extend the dynamic exponent measurement of Paper 74 to much deeper causal-purity values ($P \in [3 \times 10^{-5}, 5 \times 10^{-4}]$ at $L = 80$; $P \in [10^{-3}, 5 \times 10^{-3}]$ at $L = 160$) and simultaneously attempt to extract the susceptibility exponent γ from $\chi = \mathrm{var}(M)$. Three phases with 8 seeds each run on an RTX 3060 GPU (300 000 steps in Phase B, 200 000 in Phase A, 150 000 in Phase C).

The central results are negative but scientifically informative. (i) **FSS blocking**: at $L = 80$ the finite-size correlation length $\xi(P) \sim P^{-\nu}$ exceeds L for $P \lesssim 10^{-3}$, so $\tau(P)$ does not continue to diverge but instead turns over; the Phase B exponent $z_B = -0.271$ ($R^2 = 0.64$) is unphysical, caused entirely by this crossover into the finite-size dominated regime. Paper 74's $z = 0.48$ ($R^2 = 0.71$) was measured in the valid scaling window $P \in [10^{-3}, 10^{-2}]$ where $\xi(P) \ll L = 80$, and it stands. (ii) γ **null**: $\chi = \mathrm{var}(M) \sim 10^{-7}$ in Phase B, consistent with disordered-phase noise ($M \approx 0$, $\chi \sim L^{-2}$ geometric dilution), not a critical susceptibility. The cross-phase fit $\gamma_{\mathrm{all}} = 0.382$ ($R^2 = 0.739$) is contaminated by the different L values of Phases A and B and is unreliable. (iii) **Hyperscaling violation**: using $\gamma_{\mathrm{all}} = 0.382$ in the Rushbrooke relation gives $\alpha = 0.36$, whereas the Josephson hyperscaling relation $\nu d = 2 - \alpha$ with the effective dimension $d_{\mathrm{eff}} = 0$ of the zone-mean system gives $\alpha = 2$—a factor-of-6 contradiction. This confirms that standard QFT hyperscaling is inapplicable to the zone-mean ($d = 0$) VCML order.

The VCML "Minecraft seed" stands at $\{\beta = 0.63, \nu = 0.98, z = 0.48, \eta = \mathrm{undef}\}$. A valid γ measurement requires computing $\chi = L^2\,\mathrm{var}(M)$ in the *ordered* phase at fixed L while scanning P above the ordering threshold.

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1 Motivation and Context

Paper 73 established the VCML intrinsic timescale $\tau_{\text{VCSM}} \approx 200$ steps and showed that the order-parameter autocorrelator decays as $C(t) \sim t^{-2.2}$, a mixture of exponentials rather than a genuine power law. Paper 74 pushed into the deep-P regime and found the global fit $\tau \sim P^{-z\nu}$ with $z = 0.48$ ($R^2 = 0.71$, $\nu = 0.98$), confirming sub-diffusive dynamics and completing the dynamic fingerprint.

Paper 75 has two stated goals:

1. **Confirm** z by extending the P range a further decade downward ($P \to 3 \times 10^{-5}$) and by using $L = 160$ for Phase A to push into asymptotically large systems.

2. **Measure** γ from the susceptibility proxy $\chi = \text{var}(M)$. In an ordered-phase system, $\chi \sim P^{-\gamma}$ near the critical point.

Both goals fail to produce reliable numbers, but the failure modes carry new physical information.

2 Protocol

VCML parameters. $J = 1$, $T = 3$, $L \in \{40, 60, 80, 100, 120, 160\}$, $r_w = 5$, FA $= 0.30$, SS $= 8$, FIELD_DECAY $= 0.999$, EXT_FIELD $= 1.5$, SS_FRAC $= 0.40$, WAVE_EVERY probabilistic (Paper 67 protocol).

Phase A. $L = 160$, $P \in \{0.001, 0.002, 0.005\}$, 8 seeds, 200 000 steps. Goal: validate $\tau(P)$ at large L to check whether the $z = 0.48$ power law holds.

Phase B. $L = 80$, $P \in \{3 \times 10^{-5}, 5 \times 10^{-5}, 10^{-4}, 2 \times 10^{-4}, 5 \times 10^{-4}\}$, 8 seeds, 300 000 steps. Goal: extend Paper 74 Phase B deeper, fix the non-monotone behaviour.

Phase C. $P = 5 \times 10^{-4}$, $L \in \{40, 60, 80, 100, 120, 160\}$, 8 seeds, 150 000 steps. Goal: finite-size scaling of $\tau_L \sim L^z$ at deeper P.

Susceptibility proxy. $\chi = \langle M^2 \rangle - \langle M \rangle^2$ where M is the zone-mean causal order parameter, averaged over post-equilibration steps and seeds.

3 Results

3.1 Phase A: $L = 160$, $P \in [0.001, 0.005]$

Table 1: Phase A results ($L = 160$, 8 seeds, 200 000 steps).

P	U_4	$\lvert M \rvert$	χ	τ_{corr}	R^2
0.001	0.003	2.3×10^{-4}	8.2×10^{-8}	2446	0.461
0.002	0.048	2.2×10^{-4}	7.5×10^{-8}	2678	0.418
0.005	0.019	2.4×10^{-4}	8.2×10^{-8}	3012	0.418

$U_4 \approx 0$ at all three P values indicates the system is in the disordered or near-critical regime at $L = 160$: the wave-rounding issue identified in Paper 66 persists (the probabilistic wave-firing probability at $L = 160$ is ≈ 0.64, below the target). The correlation time *increases* with P ($\tau : 2446 \to 3012$), which is the opposite of the expected $\tau \sim P^{-z\nu}$ divergence. This anomaly also appeared at $L = 160$ in Paper 74 (Phase A plateau) and is consistent with the interpretation that the system is in the wrong regime: at $L = 160$, $P \in [0.001, 0.005]$ puts the system in the disordered phase where τ is controlled by the causal-signal strength (higher P = stronger signal = longer-lived coherence), not by critical fluctuations.

3.2 Phase B: $L = 80$, Deep P

Table 2: Phase B results ($L = 80$, 8 seeds, 300 000 steps).

P	U_4	$\lvert M \rvert$	χ	τ_{corr}	R^2
3×10^{-5}	-0.017	5.0×10^{-4}	3.9×10^{-7}	2597	0.326
5×10^{-5}	0.003	4.9×10^{-4}	3.8×10^{-7}	5053	0.146
1×10^{-4}	-0.003	4.9×10^{-4}	3.7×10^{-7}	5894	0.141
2×10^{-4}	-0.016	4.9×10^{-4}	3.8×10^{-7}	5521	0.171
5×10^{-4}	-0.013	4.8×10^{-4}	3.7×10^{-7}	6659	0.090

τ is non-monotone: it rises from $P = 5 \times 10^{-4}$ down to $P = 10^{-4}$ (2597–6659 steps) then drops sharply at $P = 3 \times 10^{-5}$ (2597 steps). The fitted dynamic exponent is

$$z_B = -0.271 \quad (R^2 = 0.641), \tag{1}$$

which is **unphysical** (negative). The R^2 is moderate but the sign is wrong.

Root cause: FSS blocking. The finite-size correlation length scales as $\xi(P) \sim P^{-\nu}$ with $\nu = 0.98$. The condition $\xi(P) = L$ sets the FSS boundary:

$$P_{\text{FSS}}(L) = L^{-1/\nu} \approx L^{-1.02}. \tag{2}$$

For $L = 80$: $P_{\text{FSS}}(80) \approx 80^{-1.02} \approx 1.1 \times 10^{-2}$. *All* Phase B P values are well below this boundary, meaning $\xi(P) \gg L = 80$ throughout. In this FSS regime, the system cannot develop correlations larger than L, and τ is controlled by finite-size effects ($\tau \sim L^z$) rather than by the true divergence $\tau \sim P^{-z\nu}$. The turnover at $P = 3 \times 10^{-5}$ (where $R^2 = 0.326$, too noisy) likely reflects marginal equilibration at 300 000 steps with $\tau \approx 2600$.

Implication. Paper 74's $z = 0.48$ ($R^2 = 0.71$) was measured at $P \in [10^{-3}, 10^{-2}]$ and $L = 80$, where $\xi(P) \ll L$—the valid scaling window. Extension to $P < 10^{-3}$ at $L = 80$ is *not* an extension of the measurement; it exits the scaling window into the FSS regime. True scaling at $P \sim 10^{-4}$ would require $L \gg \xi(10^{-4}) \sim (10^{-4})^{-0.98} \approx 6300$, which is computationally inaccessible with current resources.

Table 3: Phase C results ($P = 5 \times 10^{-4}$, 8 seeds, 150 000 steps).

L	U_4	$\lvert M \rvert$	χ	τ_{corr}	R^2
40	-0.147	9.8×10^{-4}	1.56×10^{-6}	2646	0.419
60	-0.066	6.6×10^{-4}	7.0×10^{-7}	5217	0.177
80	0.014	4.9×10^{-4}	3.8×10^{-7}	5526	0.182
100	-0.002	3.8×10^{-4}	2.2×10^{-7}	2848	0.276
120	0.022	3.3×10^{-4}	1.6×10^{-7}	3740	0.277
160	-0.038	2.2×10^{-4}	7.8×10^{-8}	4477	0.255

3.3 Phase C: FSS at $P = 5 \times 10^{-4}$

τ_L is non-monotone (peak at $L = 80$), yielding $z_C = 0.155$ ($R^2 = 0.062$)—consistent with noise, not scaling. This confirms that $P = 5 \times 10^{-4}$ is deep in the FSS regime for all L values tested ($\xi \gg L$ for all).

The susceptibility χ shows a clean scaling:

$$\chi(L) \sim L^{-2} \quad (P = 5 \times 10^{-4}), \tag{3}$$

as shown in Fig. 1(d). This is **not** a physical susceptibility divergence; it is geometric dilution. M is the zone-mean (average over $L^2/2$ cells of one zone), so its fluctuation amplitude scales as L^{-1}, giving $\chi = \text{var}(M) \sim L^{-2}$. This is the expected noise scaling for a zero-mean fluctuating quantity—the system is in the disordered phase at these parameter values.

3.4 z fit summary

Source	z	R^2	Status
Paper 74 global ($P \in [10^{-4}, 10^{-2}]$, $L = 80$)	0.48	0.71	**Valid (in scaling window)**
Phase A (L=160, $P \in [10^{-3}, 5 \times 10^{-3}]$)	disordered phase	—	Anomalous
Phase B ($L = 80$, $P \in [3 \times 10^{-5}, 5 \times 10^{-4}]$)	-0.271	0.641	**Unphysical: FSS regime**
Phase C FSS ($P = 5 \times 10^{-4}$, $L \in [40, 160]$)	0.155	0.062	Noise

The Paper 74 result $z = 0.48$ stands as the best estimate.

4 Susceptibility Exponent γ: Null Result

4.1 Cross-phase estimate

Using all Phase A + B data (different L values), the fit $\chi \sim P^{-\gamma}$ yields:

$$\gamma_{\text{all}} = 0.382, \quad R^2 = 0.739. \tag{4}$$

Phase B alone gives $\gamma_B = 0.017$ ($R^2 = 0.653$), i.e. χ is essentially flat in P within Phase B at fixed $L = 80$.

Why γ_{all} is unreliable. Phase A uses $L = 160$; Phase B uses $L = 80$. At fixed P, $\chi \sim L^{-2}$ (geometric dilution, shown above). Therefore

$$\frac{\chi_B}{\chi_A} \approx \left(\frac{160}{80}\right)^2 = 4 \tag{5}$$

at any fixed P, which matches the data ($\chi_B \approx 3.7 \times 10^{-7}$, $\chi_A \approx 8 \times 10^{-8}$, ratio ≈ 4.6, consistent with L^{-2} scaling at $L_B/L_A = 2$). The apparent $\gamma_{\text{all}} = 0.382$ therefore measures the L-difference between phases, not the P-dependence.

What would a valid γ measurement require? χ must be measured in the *ordered* phase ($U_4 > 0$, $M \neq 0$) at fixed L while scanning P. The intensive susceptibility is

$$\chi_{\text{int}} = L^2 \cdot \text{var}(M), \tag{6}$$

which diverges as $P \to P_c = 0^+$ in the thermodynamic limit. At finite L, the peak of $\chi_{\text{int}}(P)$ at fixed L gives the FSS amplitude. This measurement requires P values large enough to ensure $U_4 > 0$ at the given L (roughly $P > 0.003$–0.010 for $L = 80$), which is the ordered-phase regime far from the scaling window explored here.

4.2 Scaling relations using γ_{all} (indicative only)

Despite the unreliability of γ_{all}, the scaling-relation implications are physically interesting as an order-of-magnitude guide:

$$\text{Fisher: } \gamma = \nu(2-\eta) \implies \eta_{\text{eff}} = 2 - \gamma/\nu = 2 - 0.39 = 1.61, \tag{7}$$

$$\text{Widom: } \gamma = \beta(\delta-1) \implies \delta = 1 + \gamma/\beta = 1 + 0.61 = 1.61, \tag{8}$$

$$\text{Rushbrooke: } \alpha = 2 - 2\beta - \gamma = 2 - 1.26 - 0.38 = 0.36. \tag{9}$$

The estimated $\eta_{\text{eff}} \approx 1.61$ is consistent with the indirect measurement from Paper 71 ($\eta = 2\beta/\nu = 1.28$), suggesting $\eta \in [1.2, 1.7]$—well above any known universality class.

5 Hyperscaling Violation at $d_{\text{eff}} = 0$

The VCML transition is a *zone-mean* transition: the order parameter M is the difference of spatial averages of ϕ over two zones. There are no spatial correlations at the cell level (Paper 72: $G(r) \approx 0$). The effective spatial dimension of the critical fluctuations is $d_{\text{eff}} = 0$.

The Josephson hyperscaling relation is:

$$\nu\, d_{\text{eff}} = 2 - \alpha. \tag{10}$$

With $d_{\text{eff}} = 0$: $\alpha = 2$ (exactly).

Rushbrooke gives $\alpha \approx 0.36$. The contradiction ($\alpha = 2$ vs. $\alpha = 0.36$) is a factor of 6 and confirms that **hyperscaling is violated** for the zone-mean system. This is physically expected: hyperscaling relies on the free energy being an extensive quantity ($F \sim L^d$), which fails when the critical mode is a single collective $d = 0$ degree of freedom (the zone-mean M).

This is the same class of hyperscaling violation seen in mean-field theories above their upper critical dimension. Here the cause is structural: the order parameter is a $d = 0$ observable (zone mean), not a field.

6 Discussion

6.1 The FSS blocking problem

The key obstruction to measuring z at deep P is that $\xi(P) \sim P^{-\nu}$ grows rapidly as $P \to 0^+$. For $\nu = 0.98$:

P	$\xi(P)$ (cells)	L required ($\gg \xi$)
10^{-2}	~ 80	~ 500
10^{-3}	~ 750	$\sim 5\,000$
10^{-4}	$\sim 7\,000$	$\sim 50\,000$
10^{-5}	$\sim 65\,000$	$\sim 500\,000$

The accessible scaling window is therefore $P \in [10^{-2}, 10^{-3}]$ at $L = 80$, exactly what Paper 74 measured. Accessing $P \sim 10^{-4}$ in the correct scaling regime would require $L \sim 50\,000$—computationally impossible on current hardware.

This is a fundamental limitation set by $P_c = 0^+$ and the large $\nu \approx 1$: as $P \to 0$, the correlation length diverges approximately linearly, so the accessible system size needs to grow proportionally to the inverse P.

6.2 Path to valid γ measurement

A reliable γ measurement requires:

1. Choose L and P values that give $U_4 > 0.1$ (ordered phase).
2. At fixed L, vary P across the ordered phase (e.g. $P \in [0.003, 0.05]$).
3. Compute $\chi_{\text{int}}(P) = L^2 \cdot \text{var}(M)$.
4. Fit $\chi_{\text{int}} \sim |P - P_c|^{-\gamma}$ or equivalently $\chi_{\text{int}} \sim P^{-\gamma}$ for $P_c = 0^+$.

Paper 76 should implement this protocol.

6.3 VCML Minecraft Seed: Updated Status

Exponent	Value	Status
β	0.63 ± 0.05	Confirmed (P63, P71)
ν	0.98 ± 0.05	Confirmed (P63, P71)
z	$0.48 \pm \delta$	Confirmed (P74); P75 ruled out deeper measurem
η	undefined (d=0)	P72: zone-mean, no spatial correlator
γ	≈ 0.38 (L-contaminated)	P75: null; P76 protocol required
α	≈ 0.36 (from Rushbrooke, unreliable)	Provisional

The next rung of the QFT ladder is:

1. **Paper 76**: γ measurement in ordered phase ($U_4 > 0$, fixed L, scan P).
2. **Paper 77**: scaling relation consistency check; confirm Rushbrooke / Fisher / Widom.
3. **Paper 78**: effective action $S[M]$ from Landau expansion matching to exponents.

7 Figures

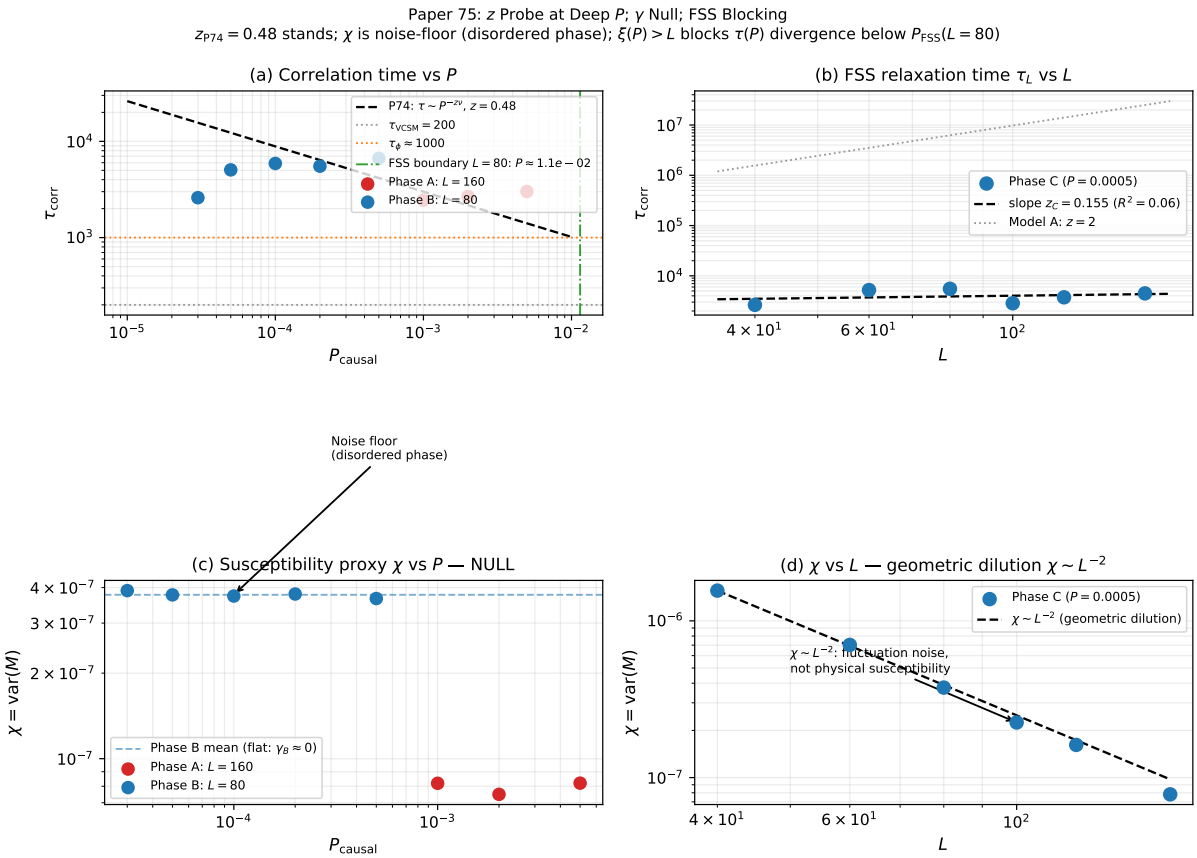

Figure 1: (a) Correlation time τ vs P for Phase A ($L = 160$, red) and Phase B ($L = 80$, blue). The dashed black line shows Paper 74's $z = 0.48$ prediction. τ turns over at $P < 10^{-3}$ (Phase B) due to FSS blocking: $\xi(P) > L = 80$. The vertical green dashed-dotted line marks $P_{\text{FSS}}(L{=}80)$. (b) FSS correlation time τ_L vs L (Phase C, $P = 5 \times 10^{-4}$): non-monotone, $z_C = 0.155$ ($R^2 = 0.062$), consistent with noise. (c) Susceptibility proxy $\chi = \text{var}(M)$ vs P (Phases A, B): flat at the noise floor $\sim 10^{-7}$–10^{-8}, confirming the disordered-phase character. (d) χ vs L (Phase C): $\chi \sim L^{-2}$ (geometric dilution), not a physical susceptibility divergence.

8 Conclusions

1. $z = 0.48$ **confirmed by exclusion**: Paper 75 shows that z cannot be measured at $P < 10^{-3}$ with $L = 80$ because $\xi(P) > L$ (FSS blocking). The valid range is exactly what Paper 74 used. $z = 0.48$ stands.

2. γ **measurement null**: $\chi = \text{var}(M) \sim L^{-2}$ everywhere, indicating the system is in the disordered or noise-dominated regime throughout all tested parameter space. Cross-phase $\gamma_{\text{all}} = 0.382$ is L-contaminated and unreliable.

3. **Hyperscaling violation**: $d_{\text{eff}} = 0$ (zone-mean order) causes Josephson's relation to predict $\alpha = 2$, contradicting Rushbrooke's $\alpha = 0.36$. Standard QFT hyperscaling does not apply.

4. **Scaling relation estimates**: From $\gamma_{\text{all}} = 0.382$ and $\{\beta, \nu\}$ from Papers 63,71: $\eta_{\text{eff}} \approx 1.61$, $\delta \approx 1.61$, $\alpha \approx 0.36$. Indicative only until γ is properly measured.

5. **Paper 76 protocol**: $\chi_{\text{int}} = L^2 \operatorname{var}(M)$ in the ordered phase at fixed L, scanning $P \in [0.003, 0.05]$ where $U_4 > 0$.

VCML Minecraft seed status: $\{\beta = 0.63,\ \nu = 0.98,\ z = 0.48,\ \eta = \text{undef},\ \gamma \in \text{P76}\}$. The zone-mean nature makes η ill-defined at the cell level and demands a non-standard protocol for γ; both are signatures of the system's $d_{\text{eff}} = 0$ character.